LF-YOLOv5: An improved and Lightweight YOLOv5 model for Disabled sign detection

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Abstract

Detection of small objects is extremely a difficult problem in computer vision research. Especially, the need for detecting and identifying the disabled car sign has become essential because of the vast increase in the vehicle numbers as well as the parking space problem. As a result, other drivers occupy the parking spots of disabled people. To avoid these circumstances, we proposed an improved and lightweight small object detection algorithm based on YOLOv5 model to detect the disabled sign on the front window of the car. We improved the existing YOLOv5 object detection model by fusing two network architectures Path Aggregation Network (PANet) and Weighted Bi-directional Feature Pyramid Network (BiFPN) which effectively increased the detection ability for small object detection algorithm. Also, by reasonably clipping the feature map output, the computing resources were significantly reduced, and the model becomes more lightweight. The experiments demonstrated that the improved algorithm has very good results in detection accuracy and real-time ability. Compared with the classical YOLOv5, the mAP@0.5 and mAP@0.5:0.95 of LF-YOLOv5 were increased by 1.4% and 0.6%, respectively, the number of parameters of the network were reduced by 68.2%, computational resources (FLOPs) were reduced by 12.7% and the inferring time was reduced by 6.7%

1. Introduction

Object detection technique is used to detect specific objects in images or video sequences by identifying their location, size, spatial relationship, and class. It has always been highly useful for research and applications in various fields, such as face recognition [1], industrial defect detection [2], traffic and vehicle detection [3], and many others. In recent years, the rise of neural networks based on convolutional, object detection algorithms using deep learning techniques have become the mainstream of contemporary object detection technology. object detection algorithms can be divided into two-stage algorithm and one-stage detection algorithm. Some them are: Faster-RCNN [4], YOLOv5 [10] , SSD [5] algorithm, etc.

The one-stage detection algorithm is gaining popularity in the field of object detection due to its high level of accuracy and speed. For early versions of object detection algorithms, the detection accuracy of the algorithm for small targets is not ideal. By adopting the image pyramid strategy [6], the above problem can be solved; however, its high computational cost and inference speed limit its implementation. Feature Pyramid Network (FPN) [7] is the mainstream configuration of object detection algorithms due to its excellent performance for small object detection when added after the backbone network. The subsequent PANet [8] and BiFPN [9] has achieved better results through various feature fusion with the basic idea of FPN.

As a result of this research, LF(Light-Fast)-YOLOv5 has been developed as an improved version of YOLOv5 for small object detection based on YOLOv5, which is able to significantly reduce the number of

parameters and calculations of the model, while having better performance in the detection of small objects than the original version.

2. Proposed Methodology

In general, the network structure of YOLOv5 refers to the backbone and neck. The backbone consists of multiple stacking of CBS(Conv + BatchNorm + SiLU) modules and C3 modules, and finally one SPPF module is connected.

2.1 Feature Map Clipping

In the multiple down sampling process of the backbone, in order to bring about the lightweightness of the algorithm, we deleted the default C5 (fifth convolution down sampling operation) layer, and made corresponding adjustments in the feature fusion part, leaving only the C3 and C4 feature maps for feature fusion. The feature map clipping corresponding to the improved method in this section is represented by the flowchart in Figure 1. The backbone has been cut accordingly to the feature map clipping and the neck represents a new feature fusion network with PBi-FPN and prediction head, which corresponds to the methods described in Section 2.2 of the improved algorithm in this paper.

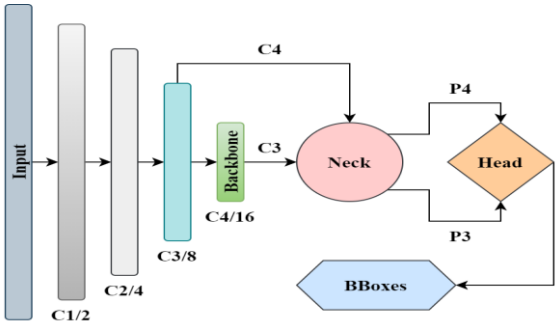


Fig 1. LF-YOLOv5 inference flowchart

2.2 Improvement of feature fusion path (PBi-FPN)

Based on the previous analysis, we designed an improved feature fusion path, as shown in Figure 2. In the part of the neck, based on PANet and BiFPN, we redesigned a new small object detection feature fusion path (PBi-FPN) as shown in figure 2(a) and (b), which aims to strengthen the fusion effect of the model on the underlying features and strengthen the small object detection performance. Meanwhile, new branches are set horizontally to participate in the fusion process of the bottom, which further enhances the detection effect of the algorithm for small targets.

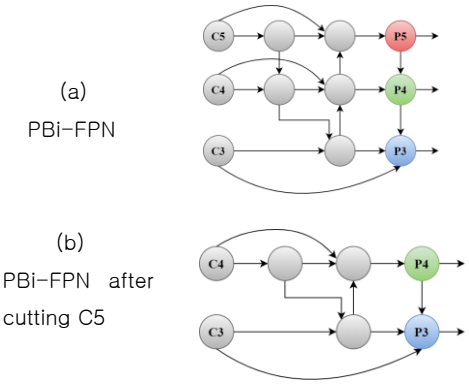


Fig 2. Proposed fusion path of PBi-FPN

3. Experimental Results

3.1Dataset

We collected a real-time dataset of 1025 car images with disabled sign and other sign sticked on the front window of the car. The images were captured using a smart phone camera with the size of 1920 x 1080 pixels. To improve the detection accuracy, we performed data augmentation techniques like tiling, flip, rotation and blur and increased the size to 15000 images.

3.2 Experimental Setup

We ran the simulation using Google colab using Python 3.8.4 version with RAM memory-12GB and a GPU backend. We split the data as train and test set in the ratio 70:30. The training epoch was set to 300.

3.3 Evaluation and analysis of experiments

At present, the mainstream general indicators for evaluating the performance of object detection algorithms include precision, recall, average precision (AP), meanAP(mAP), FLOPs (floating point operations per second), inference time, etc. As mAP more accurately reflects the algorithm's accuracy than precision or recall, for this experiment we chose mAP@0.5, mAP@0.5:0.95, parameters, FLOPs, and inference time.

A comparison of experimental performance between YOLOv5s and LF-YOLOv5 is presented in Table 1. It has been demonstrated that LF-YOLOv5 has improved comprehensive detection performance in comparison to YOLOv5s by 1.4 and 0.6, respectively, which demonstrated an improvement in mAP@0.5 and mAP@0.5:0.95. Meanwhile, the parameter (M) value for LF-YOLOv5 is decreased by 68.2% and 12.7%, respectively. Figure 3. shows the original and detection results of our proposed model with their respective confidence score.

Methods	Size	mAP @0.5	mAP @0.5 :0.95	Parameters (M)	FLOPs (G)	Time (ms)
YOLOv5s	640	69.7	35.5	7.01	15.8	13.1
LF-YOLOv5	640	71.1	36.2	2.23	13.8	12.2
Improvement	-	+1.4	+0.6	-68.2%	-12.7	-6.7%

Table 1. Performance comparison of LF-YOLOv5 and YOLOv5s



Fig 3. (a) input images and (b) output images

4. Conclusion and future works

This paper proposes a lightweight small object detection algorithm which is based on an improved feature fusion mode, and which is intended to improve the small object detection accuracy of the model. As a result, the detection performance of the algorithm is further improved and provides a good generalization ability that can be applied to scenarios involving small objects of other types of targets. In the future, we will continue to improve the detection effect of the algorithm on ultra-small targets and further explore the detection potential of the proposed algorithm.

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